

Chapter 3 — Finite elements in d dimensions

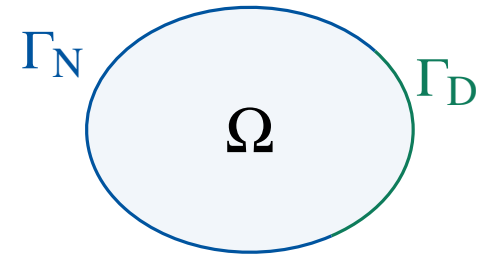
The reference element, polynomial spaces and the
geometrical map

MTH3340 · Numerical Methods for PDEs

1 The boundary value problem in d dimensions

Let $\Omega \subset \mathbb{R}^d$ ($d = 2, 3$) with boundary $\Gamma \doteq \partial\Omega$ split into a Dirichlet part Γ_D and a Neumann part Γ_N .

$$\begin{cases} \mathcal{A} u(\mathbf{x}) = f(\mathbf{x}) & \text{in } \Omega, \\ \mathcal{B}_D u(\mathbf{x}) = u_D(\mathbf{x}) & \text{on } \Gamma_D, \\ \mathcal{B}_N u(\mathbf{x}) = g_N(\mathbf{x}) & \text{on } \Gamma_N. \end{cases}$$



$\mathcal{A}$ is a differential operator (e.g. $-\Delta$), $\mathcal{B}_D$ a trace operator and $\mathcal{B}_N$ a flux operator.

Nothing here is new in kind — only in d . The whole difficulty moves to the construction of the discrete space.

The weak form

Multiply by a test function, integrate over Ω and integrate by parts. Dirichlet data are **essential** (built into the space); Neumann data are **natural** (they land in the right-hand side).

Example (Poisson problem with mixed conditions)

$-\nabla \cdot \kappa \nabla u = f$, $u = u_D$ on Γ_D , $\mathbf{n} \cdot \kappa \nabla u = g_N$ on Γ_N . Pick any extension $Eu_D \in H^1(\Omega)$ with $Eu_D = u_D$ on Γ_D . Find $u_0 \in H_{\Gamma_D}^1(\Omega)$ with

$$\int_{\Omega} \kappa \nabla u_0 \cdot \nabla v = \int_{\Omega} f v + \int_{\Gamma_N} g_N v - \int_{\Omega} \kappa \nabla [Eu_D] \cdot \nabla v \quad \forall v \in H_{\Gamma_D}^1(\Omega),$$

and set $u \doteq u_0 + Eu_D$.

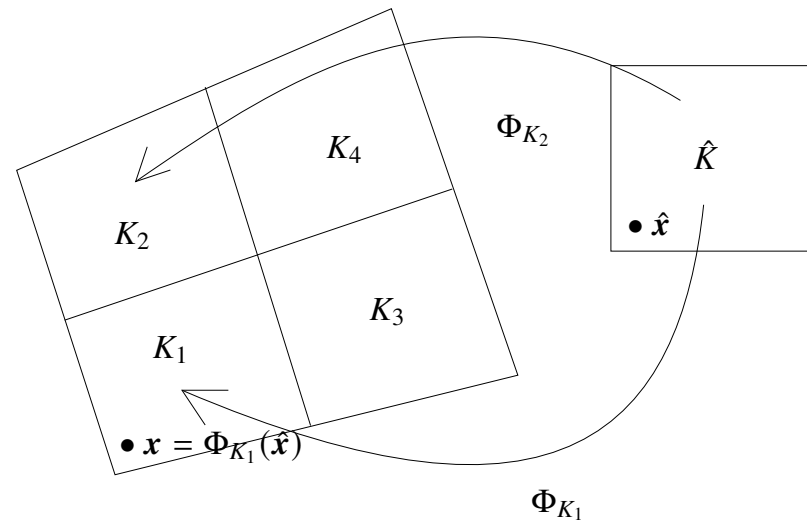
This is the **offset function method** of Chapter 2, verbatim. Abstractly: find $u \in X$ with $a(u, v) = \ell(v)$ for all $v \in X$.

Galerkin, and the one thing we are missing

Find $u_h \in X_h$ such that $a(u_h, v_h) = \ell(v_h)$ for all $v_h \in X_h$, with $X_h \subset X$ **conforming**. Céa's lemma still applies. **So: how do we build X_h in 2D and 3D?**

In 1D we wrote the hat functions down explicitly on a partition of (a, b) . In d dimensions there is no such global formula. Instead, everything is built **cell by cell** from **one reference cell $\hat{K}$** :

1. build a space on $\hat{K}$;
2. push it to each K through Φ_K ;
3. assemble, enforcing C^0 .



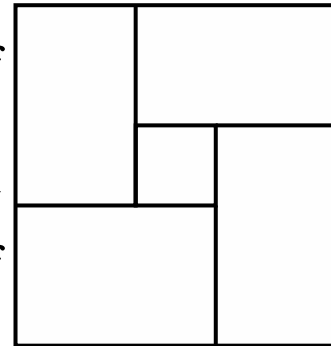
2 The mesh

Definition (Conforming triangulation)

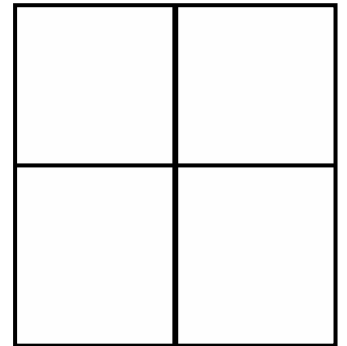
A partition $\mathcal{T}_h$ of Ω into cells K is *conforming* if for any two neighbours K^+, K^- the intersection $\overline{K}^+ \cap \overline{K}^-$ is either empty or a whole k -face ($k < d$) of both cells.

A k -face of a polytope: the cell itself is a d -face, then faces, edges, vertices.

Hanging nodes are excluded — they would break the gluing of degrees of freedom.



Non-conforming



Conforming

Every $K \in \mathcal{T}_h$ carries a reference cell $\hat{K}$ and a diffeomorphism $\Phi_K : \hat{K} \rightarrow K$; write $\hat{x} \doteq \Phi_K^{-1}(x)$.

3 The reference finite element

Definition (Finite element (Ciarlet))

A finite element is a triplet $\{K, \mathcal{V}, \Sigma\}$ where K is a compact, connected, Lipschitz subset of $\mathbb{R}^d$, $\mathcal{V}$ is a vector space of functions on K , and $\Sigma = \{\sigma_1, \dots, \sigma_{n_\Sigma}\}$ is a set of linear functionals forming a basis of the dual space $\mathcal{V}'$.

The σ_i are the degrees of freedom. Because Σ is a dual basis, there is a unique basis $\{\phi^1, \dots, \phi^{n_\Sigma}\}$ of $\mathcal{V}$ with

$$\sigma_i(\phi^j) = \delta_{ij},$$

the shape functions: one per degree of freedom. We build the triplet first on the reference cell, $(\hat{K}, \hat{\mathcal{V}}, \hat{\Sigma})$.

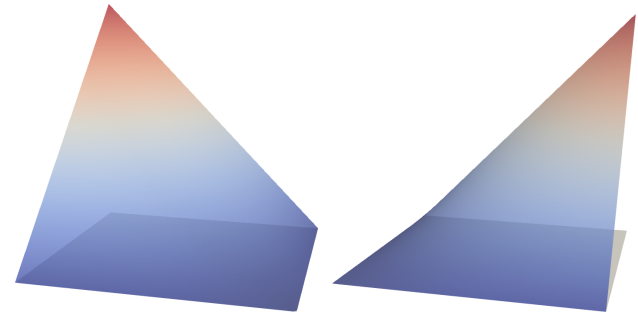
Example: the bilinear reference element

$\hat{K} \doteq [-1, 1]^2$, and

$$\begin{aligned}\hat{\mathcal{V}} &= \mathcal{P}_1(\hat{x}) \otimes \mathcal{P}_1(\hat{y}) = \{1, \hat{x}\} \otimes \{1, \hat{y}\} \\ &= \text{span}\{1, \hat{x}, \hat{y}, \hat{x}\hat{y}\} \doteq Q_1,\end{aligned}$$

with $\hat{\Sigma} = \{\hat{\sigma}_i(\hat{v}) \doteq \hat{v}(\hat{x}_i)\}$, the evaluations at the four vertices.

Is $\hat{\Sigma}$ a basis of $\hat{\mathcal{V}}$? Two bilinear polynomials with the same four nodal values are equal, so $\hat{v} \mapsto (\hat{\sigma}_1(\hat{v}), \dots, \hat{\sigma}_4(\hat{v}))$ is injective between spaces of the same dimension — hence bijective. This is unisolvence.



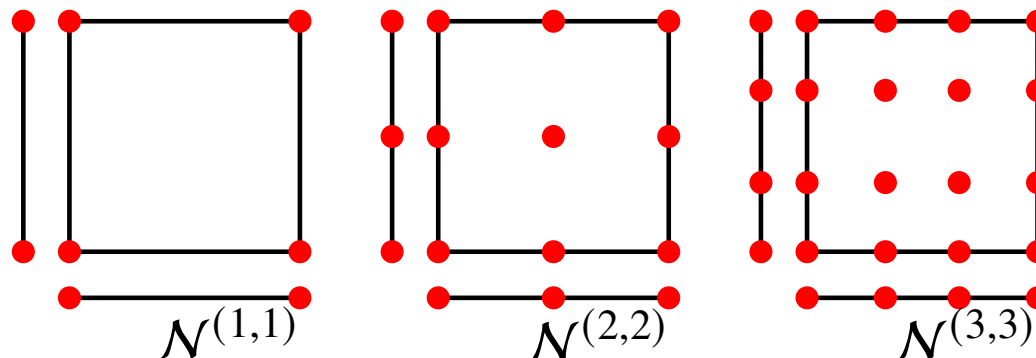
Two shape functions of Q_1 on $\hat{K}$.

Lagrangian bases and tensor-product nodes

Given nodes $\{x_0, \dots, x_k\}$ in $\mathbb{R}$, the 1D Lagrange polynomials are

$$\ell_m^k(x) \doteq \frac{\prod_{n \neq m} (x - x_n)}{\prod_{n \neq m} (x_m - x_n)}, \quad \ell_m^k(x_l) = \delta_{ml}.$$

On a d -cube we take **tensor products**, both of the nodes, $\mathcal{N}^k = \mathcal{N}^{k_1} \times \dots \times \mathcal{N}^{k_d}$, and of the polynomials, $\ell_m^k(\mathbf{x}) \doteq \prod_{i=1}^d \ell_{m_i}^{k_i}(x_i)$. E.g. in 2D for $p = 1$: $\hat{b}^1 = \ell_0(\hat{x})\ell_0(\hat{y})$, $\hat{b}^2 = \ell_1(\hat{x})\ell_0(\hat{y})$, $\hat{b}^3 = \ell_0(\hat{x})\ell_1(\hat{y})$, $\hat{b}^4 = \ell_1(\hat{x})\ell_1(\hat{y})$. So $\dim \mathcal{Q}_p = (p+1)^d$ and **the shape functions come for free**.

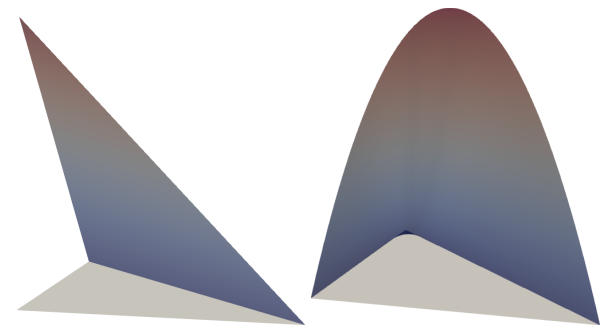
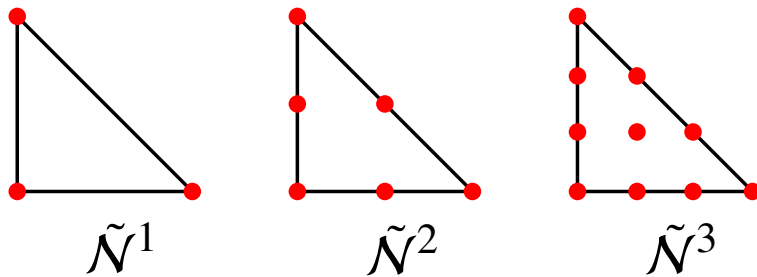


Simplices: the spaces $\mathcal{P}_p$

On triangles and tetrahedra the tensor product is **too rich**. We truncate:

$$\mathcal{P}_p \doteq \text{span}\{\mathbf{x}^\alpha : |\alpha| \leq p\}, \quad |\alpha| \doteq \sum_i \alpha_i.$$

In 2D, $\mathcal{P}_1 = \{1, x, y\}$ (dim 3) and $\mathcal{P}_2 = \{1, x, y, xy, x^2, y^2\}$ (dim 6); in general $\dim \mathcal{P}_p = \binom{p+d}{d}$. The nodes are truncated the same way, $\tilde{\mathcal{N}}^p = \{s \in \mathcal{N}^{p1} : |s| \leq p\}$ — vertices, then edge midpoints, and so on.



Shape functions of $\mathcal{P}_1$ and $\mathcal{P}_2$.

Shape functions from a pre-basis

On simplices there is no tensor-product shortcut, and for high p the formulas get unpleasant. A code should generate them. Take any pre-basis $\{\psi^1, \dots, \psi^{n_\Sigma}\}$ of $\mathcal{V}$ — monomials will do — and look for $\phi^i = \Xi_{ij}\psi^j$.

Imposing $\sigma_k(\phi^i) = \delta_{ik}$ and writing $\mathbf{C}_{kj} \doteq \sigma_k(\psi^j)$,

$$\delta_{ik} = \sigma_k(\Xi_{ij}\psi^j) = \Xi_{ij} \sigma_k(\psi^j) = \Xi_{ij} \mathbf{C}_{kj} \quad \Longrightarrow \quad \mathbf{C} \mathbf{\Xi}^T = \mathbf{I}, \quad \mathbf{\Xi} = \mathbf{C}^{-T}.$$

One small dense solve, once, on the reference cell. It also tells us when the element is well defined: $\mathbf{C}$ is invertible exactly when Σ is unisolvent on $\mathcal{V}$.

4 From the reference to the physical cell

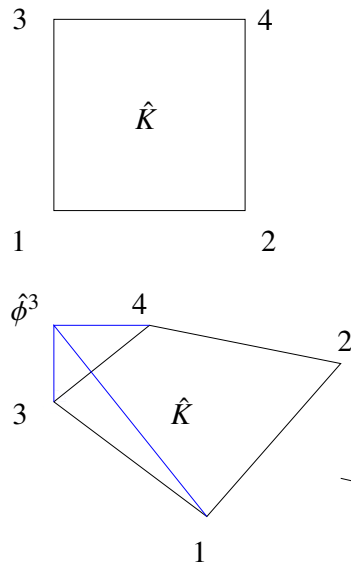
Given $(\hat{K}, \hat{\mathcal{V}}, \hat{\Sigma})$ and the map Φ_K , the finite element $(K, \mathcal{V}, \Sigma)$ on the physical cell is

$$K \doteq \Phi_K(\hat{K}), \quad \mathcal{V} \doteq \{\hat{v} \circ \Phi_K^{-1} : \hat{v} \in \hat{\mathcal{V}}\}, \quad \Sigma \doteq \{\sigma_i(v) \doteq \hat{\sigma}_i(v \circ \Phi_K)\}.$$

We never build a space on K from scratch: everything is $\hat{K}$ plus a map.

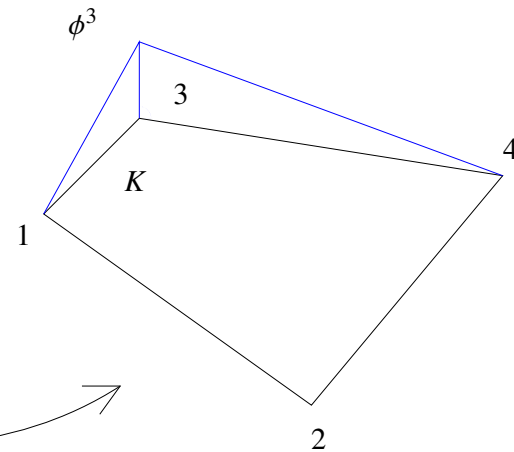
$$\hat{\mathcal{V}} \doteq \text{span}\{\hat{b}^1, \hat{b}^2, \hat{b}^3, \hat{b}^4\}$$

$$\doteq \text{span}\{1, \hat{x}, \hat{y}, \hat{x}\hat{y}\}$$



$$\mathcal{V} \doteq \text{span}\{b^1, b^2, b^3, b^4\},$$

$$\text{where } b^i(x) \doteq \hat{b}^i \circ \Phi_K^{-1}(x)$$



Degrees of freedom and shape functions on K

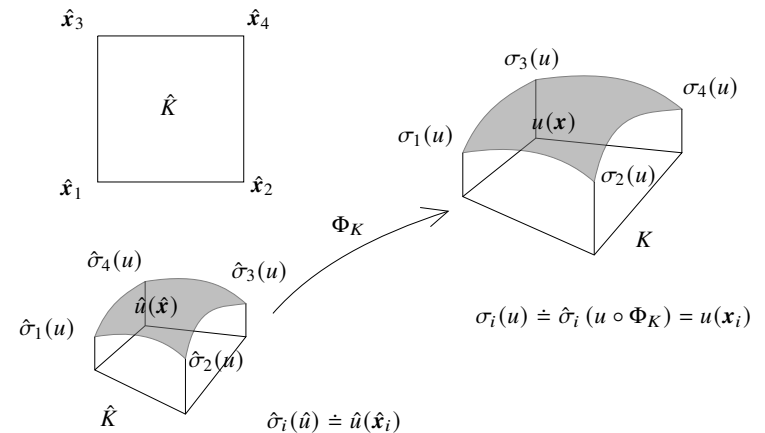
For Lagrangian elements the pull-back is transparent,

$$\sigma_i(v) = \hat{\sigma}_i(v \circ \Phi_K) = v(\mathbf{x}_i),$$

just the nodal value at the mapped node. And the duality survives the map: with $\phi^j \doteq \hat{\phi}^j \circ \Phi_K^{-1}$,

$$\sigma_i(\phi^j) = \hat{\sigma}_i(\hat{\phi}^j) = \delta_{ij}.$$

So the local interpolator $\pi_K(v) \doteq \sum_i \sigma_i(v) \phi^i$ is a projection onto $\mathcal{V}$.



5 How do we compute Φ_K ?

The mesh gives us the vertex positions $\mathbf{x}^i$ of each cell. Use the finite element machinery on the geometry itself:

$$\mathbf{x} = \Phi_K(\hat{\mathbf{x}}) \doteq \sum_i \hat{\phi}_1^i(\hat{\mathbf{x}}) \mathbf{x}^i,$$

with $\{\hat{\phi}_1^i\}$ the *linear* reference shape functions. Since $\hat{\phi}_1^i(\hat{\mathbf{x}}_j) = \delta_{ij}$, the map sends each reference vertex to the corresponding physical vertex, as required.

Triangle ($\hat{\mathcal{V}} = \mathcal{P}_1$): three vertices determine $\Phi_K = \mathbf{A}\hat{\mathbf{x}} + \mathbf{b}$, **affine**, so $\mathbf{J}_K = \mathbf{A}$ is *constant* on the cell and $\Phi_K^{-1}(\mathbf{x}) = \mathbf{A}^{-1}\mathbf{x} - \mathbf{A}^{-1}\mathbf{b}$.

Quadrilateral ($\hat{\mathcal{V}} = \mathcal{Q}_1$): the $\hat{x}\hat{y}$ term survives, so Φ_K is **bilinear**, **not affine**, and $\mathbf{J}_K$ varies within the cell.

Which space do we actually get on K ?

Triangles. $\hat{v} = a + b\hat{x} + c\hat{y}$ and $\hat{x} = \tilde{A}\mathbf{x} + \tilde{\mathbf{b}}$ is affine, so

$$v = \hat{v} \circ \Phi_K^{-1} = \tilde{a} + \tilde{b}x + \tilde{c}y \in \mathcal{P}_1(K). \quad \checkmark$$

Affine maps preserve $\mathcal{P}_p$.

General quadrilaterals. $\hat{v} \in Q_1(\hat{K})$ but $v = \hat{v} \circ \Phi_K^{-1} \notin Q_1(K)$ — the map is not affine, so v is not even a polynomial. Equality holds only for an **axis-aligned rectangle** (stretch + translation of $\hat{K}$): a rotation or a shear turns $\hat{x}\hat{y}$ into a quadratic with x^2, y^2 terms. In every case $\mathcal{P}_1(K) \subset \text{span}\{\phi^i\}$.

